ETC3410: Applied Econometrics
Assignment 1

Zhiqi Fang
Student ID: 35954396

Question 1: Binary Outcome Model

1. Describing the data

(a) Create a binary variable

A binary indicator can be defined as

$$parr86_i = \begin{cases} 1, & \text{if } nparr86_i > 0, \\ 0, & \text{if } nparr86_i = 0. \end{cases}$$

Accordingly, $parr86_i$ takes the value 1 when individual i was arrested for a property crime at least once in 1986, and 0 otherwise.

(b) Fraction arrested for property crimes in 1986

Since $parr86_i$ is a binary variable, its sample mean is equal to the proportion of individuals in the sample with $parr86_i = 1$. Therefore, the fraction of the sample arrested for property crimes in 1986 is

$$\overline{parr86} = \frac{1}{n}\sum_{i=1}^{n} parr86_i = 0.093945$$

Hence, approximately 9.39% of the sample was arrested for a property crime in 1986.

(c) Proportions among those arrested for property crimes in 1986

Since $black$, $hispan$, and $born60$ are binary variables, their conditional means are equal to the corresponding conditional probabilities.

(i) Black

$$P(black = 1 \mid parr86 = 1) = E(black \mid parr86 = 1) = 0.28125.$$

Thus, about 28.13% of those arrested for property crimes in 1986 are Black.

(ii) Hispanic

$$P(hispan = 1 \mid parr86 = 1) = E(hispan \mid parr86 = 1) = 0.2109375$$

Thus, about 21.09% of those arrested for property crimes in 1986 are Hispanic.

(iii) Born in 1960

$$P(born60 = 1 \mid parr86 = 1) = E(born60 \mid parr86 = 1) = 0.3476563$$

Thus, about 34.77% of those arrested for property crimes in 1986 were born in 1960.

(d) Average prior conviction rate

The average prior conviction rate is the conditional mean of $pcnv$. For individuals arrested for property crimes in 1986,

$$E(pcnv \mid parr86 = 1) = 0.3199219$$

while for those not arrested,

$$E(pcnv \mid parr86 = 0) = 0.3617132.$$

Thus, the average prior conviction rate is about 31.99% for those arrested for property crimes in 1986 and 36.17% for those not arrested. Therefore, the average prior conviction rate is slightly lower among those arrested for property crimes in 1986 than among those not arrested.

2. Linear Probability and Logit models

(a) LPM of $parr86$ on $pcnv$

$$parr86_i = \beta_0 + \beta_1 pcnv_i + u_i \tag{1}$$

The estimated simple linear probability model is

$$\begin{array}{rccccl} \widehat{parr86}_i = & \underset{(0.0074)}{0.1021} & - & \underset{(0.0121)}{0.0228} & pcnv_i \end{array} \tag{2}$$

(b) Interpretation

Interpretation: The estimated coefficient on *pcnv* is -0.0228. In the LPM, this implies that a one-unit increase in the proportion of prior arrests leading to conviction reduces the predicted probability of being arrested for a property crime in 1986 by 2.28 percentage points.

Expected relationship: Yes, the negative estimate is consistent with economic intuition. A higher share of previous arrests leading to conviction increases the expected cost of committing crime, which may discourage future offending. It may also reflect an incapacitation effect, since higher conviction rates can mean more time in prison and therefore fewer opportunities to be arrested for a property crime in 1986.

Statistical significance: At the 5% level, the estimate for *pcnv* is not statistically significant, since its robust p-value is 0.060, which is greater than 0.05. Its 95% confidence interval also contains zero, so we do not reject the null hypothesis that the coefficient on *pcnv* equals zero. Nevertheless, the coefficient is significant at the 10% level, suggesting only weak evidence that *pcnv* is negatively associated with the probability of being arrested for a property crime in 1986.

(c) Logit model of $parr86$ on $pcnv$

Logit results The estimated logit model is

$$P(parr86_i = 1 \mid pcnv_i) = \Lambda(\beta_0 + \beta_1 pcnv_i), \tag{3}$$

where $\Lambda(z) = \dfrac{e^z}{1+e^z}$. Using the estimation results, the fitted model is

$$P(parr86 = 1 \mid pcnv) = \Lambda\left(\underset{(0.0860)}{-2.1726} - \underset{(0.1713)}{0.2755}\, pcnv\right) \tag{4}$$

Sign and statistical significance Yes, the estimated coefficient on *pcnv* remains negative, which is consistent with part (b). However, it is not statistically significant at the 5% level because the p-value is 0.108, and it is not significant at the 10% level either.

Thus, although the direction of the effect is unchanged, the evidence is weaker than in part (b), where *pcnv* was significant at the 10% level.

Marginal effect of *pcnv* In the logit model, the marginal effect of *pcnv* is

$$\frac{\partial P(parr86 = 1 \mid pcnv)}{\partial pcnv} = \Lambda(\beta_0 + \beta_1 pcnv)\big[1 - \Lambda(\beta_0 + \beta_1 pcnv)\big]\beta_1 \tag{5}$$

Using the estimated coefficients, this becomes

$$\frac{\partial P(parr86 = 1 \mid pcnv)}{\partial pcnv} = \Lambda(-2.1726-0.2755\, pcnv)\big[1-\Lambda(-2.1726-0.2755\, pcnv)\big](-0.2755)$$

Comparison with the LPM In the LPM from part (a), the marginal effect is simply

$$\frac{\partial P(parr86 = 1 \mid pcnv)}{\partial pcnv} = \beta_1 = -0.0228$$

By contrast, in the linear probability model, the marginal effect is simply β_1. Therefore, the LPM has a constant marginal effect, while the logit model has a nonlinear marginal effect that depends on the covariates and thus varies across individuals.

(d) LPM with additional controls

Estimated equation Let X_i collect the remaining control variables ($avgsen_i$, $tottime_i$, $ptime86_i$, $qemp86_i$, $inc86_i$, $black_i$, $hispan_i$, $born60_i$). Then the estimated model can be written as

$$\begin{aligned}\widehat{parr86}_i = \underset{(0.0138)}{0.1199} + \underset{(0.0595)}{0.2286}\; pcnv_i - \underset{(0.0587)}{0.2574}\; pcnv_i^2 + X_i\hat{\gamma},\end{aligned} \tag{6}$$

where

$$\begin{aligned}X_i\hat{\gamma} = {} & \underset{(0.0033)}{-0.0044}\; avgsen_i + \underset{(0.0030)}{0.0008}\; tottime_i - \underset{(0.0022)}{0.0087}\; ptime86_i - \underset{(0.0050)}{0.0050}\; qemp86_i \\ & - \underset{(0.000087)}{0.00053}\; inc86_i + \underset{(0.0191)}{0.0675}\; black_i + \underset{(0.0134)}{0.0051}\; hispan_i - \underset{(0.0115)}{0.0013}\; born60_i.\end{aligned}$$

Marginal effect of *pcnv* Since the model includes both *pcnv* and $pcnv^2$, the marginal effect of $pcnv_i$ is

$$\frac{\partial \widehat{parr86}_i}{\partial pcnv_i} = \hat{\beta}_1 + 2\hat{\beta}_2\, pcnv_i. \tag{7}$$

Using the estimates,

$$\frac{\partial \widehat{parr86}_i}{\partial pcnv_i} = 0.2286 + 2(-0.2574)\, pcnv_i = 0.2286 - 0.5149\, pcnv_i.$$

Thus, the marginal effect of $pcnv_i$ is not constant.

Average marginal effect The average marginal effect is

$$AME = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\beta}_1 + 2\hat{\beta}_2\, pcnv_i\right) = 0.0444. \tag{8}$$

Hence, on average, a one-unit increase in pcnv increases the predicted probability of arrest for a property crime in 1986 by about 4.44 percentage points, holding the other variables fixed.

Statistical significance Using the delta-method output from `margins`, the estimated average marginal effect has robust standard error 0.0201, $t = 2.21$, and $p = 0.027$, with a 95% confidence interval of $[0.0050,\ 0.0839]$. Since $p = 0.027 < 0.05$, the average marginal effect of `pcnv` is statistically significant at the 5% level.

(e) Logit model with additional controls

Estimated model The estimated logit model is

$$P(parr86_i = 1 \mid X_i) = \Lambda\left(\begin{array}{ccccc} -2.0608 & + & 2.4009 & pcnv_i - & 2.9234 \quad pcnv_i^2 + X_i\hat{\gamma} \\ (0.1559) & & (0.6119) & & (0.6533) \end{array}\right), \tag{9}$$

where $\Lambda(z) = \dfrac{e^z}{1+e^z}$, and

$$\begin{aligned} X_i\hat{\gamma} = \; & \underset{(0.0646)}{-0.0666}\; avgsen_i + \underset{(0.0452)}{0.0150}\; tottime_i - \underset{(0.0426)}{0.0937}\; ptime86_i + \underset{(0.0663)}{0.1144}\; qemp86_i \\ & - \underset{(0.0028)}{0.0166}\; inc86_i + \underset{(0.1644)}{0.6105}\; black_i + \underset{(0.1752)}{0.0652}\; hispan_i - \underset{(0.1423)}{0.0108}\; born60_i. \end{aligned}$$

Marginal effect of ***pcnv*** Since the model includes both $pcnv$ and $pcnv^2$, the marginal effect of $pcnv_i$ in the logit model is

$$\frac{\partial P(parr86_i = 1 \mid X_i)}{\partial pcnv_i} = \Lambda(z_i)\big[1 - \Lambda(z_i)\big]\left(\beta_1 + 2\beta_2\, pcnv_i\right), \tag{10}$$

where

$$z_i = \beta_0 + \beta_1\, pcnv_i + \beta_2\, pcnv_i^2 + X_i\gamma.$$

Using the estimated coefficients, this becomes

$$\frac{\partial P(parr86_i = 1 \mid X_i)}{\partial pcnv_i} = \Lambda(\hat{z}_i)\big[1 - \Lambda(\hat{z}_i)\big]\,(2.4009 - 5.8469\, pcnv_i)\,.$$

Thus, the marginal effect of $pcnv_i$ is not constant; it depends both on the logistic density term and on the value of $pcnv_i$.

Average marginal effect The estimated average marginal effect is

$$AME(pcnv) = 0.0425 \tag{11}$$

With a standard error of 0.0155, the estimate suggests that a one-unit increase in *pcnv* raises the predicted probability of a property-crime arrest in 1986 by approximately 4.25 percentage points, holding all other variables constant.

Statistical significance Yes, since the p-value is 0.006 less than 0.05, the average marginal effect is statistically significant at the 5% level. Equivalently, the 95% confidence interval does not include zero.

(f) Marginal effects

(i) Using the LPM in part (d) From the linear probability model in part (d),

$$P(parr86_i = 1 \mid X_i) = \beta_0 + \beta_1 pcnv_i + \beta_2 pcnv_i^2 + X_i\gamma \tag{12}$$

Holding the other regressors fixed, the estimated change in the probability of property-crime arrest when *pcnv* increases from 0.20 to 0.25 is

$$\Delta\hat{P} = \hat{\beta}_1(0.25 - 0.20) + \hat{\beta}_2(0.25^2 - 0.20^2) \tag{13}$$

Using $\hat{\beta}_1 = 0.2286$ and $\hat{\beta}_2 = -0.2574$,

$$\Delta\hat{P} = 0.2286(0.05) - 0.2574(0.0625 - 0.04) = 0.00564$$

Hence, increasing *pcnv* from 0.20 to 0.25 raises the predicted probability of property-crime arrest by about 0.00564, that is, by about 0.56 percentage points.

(ii) Using the logit model in part (e) Using the logit model in part (e), I evaluate the predicted probability of property-crime arrest at $pcnv = 0.20$ and $pcnv = 0.25$,

holding *avgsen*, *tottime*, *ptime*86, *qemp*86, and *inc*86 at their sample means, and setting $hispan = 1$ and $born60 = 1$.
The predicted probabilities are

$$\hat{P}(parr86 = 1 \mid pcnv = 0.20) = 0.0946$$

and

$$\hat{P}(parr86 = 1 \mid pcnv = 0.25) = 0.0994$$

Thus, the estimated effect of increasing *pcnv* from 0.20 to 0.25 is

$$\widehat{\Delta P} = 0.0994 - 0.0946 = 0.0048. \tag{14}$$

Using the exact `lincom` result, $\widehat{\Delta P} = 0.0047523$
Thus, for a Hispanic person born in 1960, increasing *pcnv* from 0.20 to 0.25 raises the predicted probability of a property-crime arrest by about 0.48 percentage points. This is close to the LPM estimate of 0.56 percentage points and is statistically significant at the 5% level ($z = 2.74,\ p = 0.006$).

(iii) Average marginal effect of being black Because *black* is a binary variable, its average marginal effect is interpreted as an average discrete change rather than a derivative. In the logit model from part (e), the average marginal effect of being black is

$$AME(black) = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{P}_i(black_i = 1) - \hat{P}_i(black_i = 0)\right] \tag{15}$$

Equivalently,

$$AME(black) = \frac{1}{n}\sum_{i=1}^{n}\left[\Lambda(\hat{z}_i \mid black_i = 1) - \Lambda(\hat{z}_i \mid black_i = 0)\right],$$

where

$$\Lambda(z) = \frac{e^z}{1+e^z}.$$

From the `margins, dydx(black)` output, the estimated average marginal effect is

$$AME(black) = 0.0493.$$

Thus, on average, being Black increases the predicted probability of property-crime arrest by about 4.93 percentage points, holding the other variables fixed. This effect is statistically significant, since the z-statistic is 3.71, the p-value is 0.000, and the 95% confidence interval $[0.0233,\ 0.0754]$ does not include zero.

(g) LR test for joint significance

We test whether *avgsen*, *tottime*, and *inc*86 are jointly significant in the logit model from part (e).

Hypotheses The null hypothesis is

$$H_0: \ \beta_{avgsen} = \beta_{tottime} = \beta_{inc86} = 0,$$

against the alternative

$$H_1: \ \text{at least one of } \beta_{avgsen}, \beta_{tottime}, \beta_{inc86} \neq 0.$$

Thus, under the null hypothesis, *avgsen*, *tottime*, and *inc*86 are jointly insignificant in the logit model.

Restricted and unrestricted models The unrestricted model is the same as the logit model in part (e):

$$\log\left(\frac{P(parr86_i = 1 \mid X_i)}{1 - P(parr86_i = 1 \mid X_i)}\right) = \beta_0 + \beta_1 pcnv_i + \beta_2 pcnv_i^2 + X_i\gamma, \tag{16}$$

where

$$X_i = (avgsen_i,\ tottime_i,\ ptime86_i,\ qemp86_i,\ inc86_i,\ black_i,\ hispan_i,\ born60_i).$$

Its log-likelihood is

$$\ell_U = -777.09769.$$

The restricted model imposes

$$\beta_{avgsen} = \beta_{tottime} = \beta_{inc86} = 0,$$

so it excludes these three variables:

$$\begin{aligned}\log\left(\frac{P(parr86_i = 1 \mid X_i)}{1 - P(parr86_i = 1 \mid X_i)}\right) &= \beta_0 + \beta_1 pcnv_i + \beta_2 pcnv_i^2 + \beta_5 ptime86_i \\ &\quad + \beta_6 qemp86_i + \beta_8 black_i + \beta_9 hispan_i + \beta_{10} born60_i.\end{aligned} \tag{17}$$

Its log-likelihood is

$$\ell_R = -802.56165.$$

Test statistic The likelihood-ratio statistic is

$$LR = 2(\ell_U - \ell_R). \tag{18}$$

Substituting the estimated log-likelihood values,

$$LR = 2\left[(-777.09769) - (-802.56165)\right] = 2(25.46396) = 50.92792 \approx 50.93. \tag{19}$$

Under H_0, the LR statistic has an asymptotic chi-squared distribution with degrees of freedom equal to the number of restrictions. Since there are three restrictions,

$$LR \sim \chi^2(3) \quad \text{under } H_0. \tag{20}$$

Decision rule At the 5% significance level, reject H_0 if

$$LR > \chi^2_{0.95}(3) = 7.815. \tag{21}$$

Equivalently, reject H_0 if the p-value is less than 0.05.
From the output,

$$LR\chi^2(3) = 50.93, \qquad p\text{-value} = 0.0000.$$

Since $50.93 > 7.815$ and $0.0000 < 0.05$, we reject H_0.

Conclusion There is very strong evidence that $avgsen$, $tottime$, and $inc86$ are jointly significant in the logit model in part (e). Therefore, these variables should not be omitted jointly from the model.

Question 2

2.1 Definitions

(a) In a binary response model, the response probability is the probability that the dependent variable takes the value one, conditional on the explanatory variables:

$$P(y_i = 1 \mid X_i).$$

It represents the likelihood of the outcome of interest for an individual with characteristics X_i.

(b) The linear probability model (LPM) takes the form

$$P(y_i = 1 \mid X_i) = \beta_0 + X_i\beta.$$

Its main drawbacks are that predicted probabilities may fall outside the interval $[0, 1]$, the disturbance term is heteroskedastic, and the marginal effect of each explanatory variable is assumed to be constant. Hence, although the LPM is easy to estimate and interpret, it may not be well suited to binary outcomes.

(c) In binary choice models, the response probability is expressed as

$$P(y_i = 1 \mid X_i) = G(X_i\beta),$$

where $G(\cdot)$ denotes a cumulative distribution function (CDF). A suitable CDF must map any real number into the interval $(0, 1)$, ensuring that predicted probabilities are valid. It should also be nondecreasing and smooth, so that probabilities change in a sensible way and marginal effects are well defined.

(d) In a logit model, the probability is written as

$$P(y_i = 1 \mid X_i) = \Lambda(X_i\beta),$$

so the marginal effect of a continuous regressor generally differs across observations. The average marginal effect (AME) is the average of these individual marginal effects over the sample:

$$AME_k = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial P(y_i = 1 \mid X_i)}{\partial x_{ki}}.$$

Thus, the AME measures the average change in the predicted probability associated with a one-unit rise in x_{ki}.

2.2 Testing linear restrictions

Consider the linear regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + u_i \tag{22}$$

We want to test the following set of null hypotheses simultaneously:

$$\text{Hypothesis 1: } \beta_1 + \beta_2 + \beta_3 + \beta_4 = -5$$

$$\text{Hypothesis 2: } \beta_4 = -2$$

$$\text{Hypothesis 3: } \beta_1 - \beta_2 = 0$$

$$\text{Hypothesis 4: } \beta_4 - 3\beta_3 = 0$$

(a) In this linear regression model, the marginal effect of each regressor on y is given by its slope coefficient. Therefore, the four hypotheses can be stated as linear restrictions on the parameter vector.
Let

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix}.$$

Then the joint null hypothesis can be written as

$$H_0 : R\beta = r, \tag{23}$$

where

$$R = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix}, \qquad r = \begin{bmatrix} -5 \\ -2 \\ 0 \\ 0 \end{bmatrix}. \tag{24}$$

The alternative hypothesis is

$$H_1 : R\beta \neq r,$$

which means that at least one of the restrictions is violated.

(b) The Wald statistic for testing $H_0 : R\beta = r$ is

$$W = (R\hat{\beta} - r)' \left[R\widehat{\mathrm{Var}}(\hat{\beta})R' \right]^{-1} (R\hat{\beta} - r), \tag{25}$$

where $\hat{\beta}$ is the estimator of β, and $\widehat{\mathrm{Var}}(\hat{\beta})$ is the estimated covariance matrix of $\hat{\beta}$. This statistic captures how far the estimated coefficients are from satisfying the restrictions, after accounting for estimation uncertainty.

(c) Under H_0, the Wald statistic is asymptotically chi-squared with degrees of freedom equal to the number of restrictions. Since there are four restrictions,

$$W \overset{a}{\sim} \chi^2(4). \tag{26}$$

(d) At the 5% significance level, reject H_0 if

$$W > \chi^2_{0.95}(4) = 9.488. \tag{27}$$

Equivalently, reject H_0 whenever the p-value is below 0.05.